

Ricardo (Zhicong) Xie

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: <https://zxie588-alt.github.io> | GitHub: github.com/zxie588-alt

Mechanical / Building Services Intern | HVAC, MVHR/ERV and CAD Documentation

PROFILE

- Mechanical engineering master's student in Auckland with a New Zealand-focused MVHR/ERV project covering layout, airflow schedule, duct velocity, pressure-loss estimate, equipment selection and commissioning checks.
- Interested in building services, HVAC/MEP, sustainability, facilities and consulting internships where drawings, assumptions and clear communication matter.
- Project screenshots and source links: <https://zxie588-alt.github.io>

WORK RIGHTS AND AVAILABILITY

- NZ student visa. Eligible to work up to 25 hours/week during semester and full-time during scheduled university breaks.
- Available full-time from 17 Nov 2026 to late Feb 2027. Auckland preferred; open to NZ relocation for the right internship opportunity.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study/projects in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Built a residential MVHR/ERV concept around a 3D duct layout, airflow schedule, duct velocity checks and preliminary pressure-loss estimate.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Used 60 L/s balanced supply/extract as the normal case and an 85 L/s boost extract check; estimated about 80 Pa normal and 120 Pa boost external static pressure.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; added a commissioning test plan using CO2, humidity, airflow and fan-power measurements.

Solar Chimney CFD Reproduction

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD reproduction workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Checked heat input, mesh sensitivity, mass balance, heat balance, Reynolds/Rayleigh scaling and solver behaviour before trusting the velocity results.
- Used the project to practise saying clearly what a model can support, what it cannot, and what physical data would be needed next.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering side of a team concept for a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Pulled subsystem work into one decision story across orbit/coverage, power, thermal rejection, communications capacity, launch mass, cost and sustainability.

Manufacturing Workshop Training

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and process safety.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Acted as a small-group lead during automotive structure training, explaining seat/interior structures and vehicle-system context to peers.

TECHNICAL SKILLS

- Building services: MVHR/ERV layout, airflow scheduling, duct velocity checks, pressure-loss estimate, fan-duty assumptions, equipment selection
- NZ context: NZBC G4/H1, Healthy Homes ventilation checks, commissioning and sensor-validation planning
- CAD/documentation: SolidWorks, AutoCAD/DXF, component lists, colour-coded layouts, project screenshots and portfolio documentation
- Analysis: ANSYS Fluent/CFX, heat transfer, natural convection, mass/heat balance checks
- Communication: requirements, assumptions, trade-offs, visual explanation and client/team documentation

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Former student news-department lead with photography/event documentation experience; comfortable turning technical work into clear visual evidence.
- Professional working proficiency in English; native Mandarin Chinese.